

BEDFORD 5 WNW, IN, USA

WMO: 724310

Lat: 38.8882N

Lon: 86.5707W

Elev: 760

StdP: 14.30

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 63898

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.8	(c) 9.9	(d) -3.0	(e) 4.8	(f) 6.4	(g) 1.6	(h) 6.1	(i) 11.9	(j) 12.9	(k) 33.5	(l) 11.9	(m) 29.9	(n) 3.8	(o) N/A	(p) 0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	90.1	75.3	87.7	74.6	85.7	74.0	78.5	86.2	77.2	84.7	75.8	83.1	4.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	141.0	83.7	74.9	134.6	82.0	73.5	128.3	80.2	42.6	86.1	41.1	84.9	39.8	83.1	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.1	(b) 9.7	(c) 8.5	(e) -1.7	(f) 92.6	(g) 6.4	(h) 3.9	(i) -6.3	(j) 95.4	(k) -10.0	(l) 97.6	(m) -13.6	(n) 99.8	(o) -18.3	(p) 102.6		
(a) 11.1	(b) 9.7	(c) 8.5	(e) -1.9	(f) 80.7	(g) 6.7	(h) 1.4	(i) -6.7	(j) 81.7	(k) -10.6	(l) 82.6	(m) -14.4	(n) 83.4	(o) -19.2	(p) 84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	29.7	34.4	44.6	55.8	64.6	72.6	74.8	73.7	68.2	56.7	44.1	35.9		
	DBStd	18.14	12.07	12.33	11.17	9.02	8.05	5.26	5.24	4.70	6.94	9.22	9.77	10.45		
	HDD50	2095	637	455	241	46	2	0	0	0	0	37	226	451		
	HDD65	4869	1095	856	637	294	111	8	2	3	45	290	627	901		
	CDD50	3808	7	19	73	220	454	678	768	734	547	243	51	14		
	CDD65	1109	0	1	4	19	98	236	305	271	142	32	1	0		
	CDH74	8756	0	1	18	125	696	1850	2586	2110	1151	215	4	0		
	CDH80	2608	0	0	0	5	119	552	881	645	364	42	0	0		
(14) Wind	WSAvg	3.8	4.9	5.0	5.2	4.9	3.3	2.8	2.6	2.5	2.6	3.4	4.0	4.4		
(15) Precipitation	PrecAvg	44.6	2.8	2.6	3.9	4.2	5.2	4.2	4.4	3.9	2.9	3.2	3.7	3.3		
	PrecMax	61.5	8.0	6.0	10.9	13.0	11.1	10.2	10.6	8.4	7.5	10.6	8.6	6.9		
	PrecMin	32.2	0.4	0.2	1.0	1.1	1.9	0.4	0.6	0.3	0.1	0.4	1.0	0.6		
	PrecStd	6.8	1.7	1.4	2.2	2.3	2.4	2.0	2.3	2.1	1.9	2.1	1.8	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.4	69.1	76.9	80.5	86.4	90.7	97.3	92.7	90.8	84.9	72.8	66.2		
		MCWB	59.2	59.7	63.4	63.8	73.0	74.8	74.9	74.6	73.7	69.3	60.8	60.2		
	2%	DB	57.9	63.5	71.4	77.8	83.2	87.8	90.4	89.0	87.6	79.6	68.0	61.3		
		MCWB	53.3	56.4	59.5	62.6	71.0	74.1	76.7	74.2	72.5	68.0	59.1	57.4		
	5%	DB	53.3	58.6	66.5	74.8	80.9	85.9	87.5	86.1	84.6	75.6	63.7	56.3		
		MCWB	49.1	53.3	56.1	61.3	69.3	73.9	76.6	73.8	72.2	64.7	55.7	52.3		
	10%	DB	48.0	53.6	61.9	71.3	78.3	83.3	84.8	83.8	80.8	71.9	59.6	51.2		
		MCWB	44.1	47.9	53.4	59.5	67.9	72.4	75.0	72.8	70.1	63.2	52.0	47.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	61.8	64.6	68.1	74.9	78.8	80.8	79.8	76.8	73.2	64.7	61.5		
		MCDB	62.1	68.5	75.7	76.8	83.0	86.6	88.7	88.9	86.1	82.1	70.0	65.1		
	2%	WB	54.0	58.4	61.4	65.5	72.9	77.2	79.2	77.5	75.0	70.4	60.7	57.5		
		MCDB	56.4	62.2	69.3	73.8	81.0	85.3	86.8	85.7	84.2	75.9	66.3	60.5		
	5%	WB	50.0	54.4	58.3	63.3	71.3	75.5	77.7	76.1	73.5	67.4	56.7	53.1		
		MCDB	52.3	58.3	64.7	70.1	78.9	83.1	84.9	83.2	81.9	73.0	62.0	56.2		
	10%	WB	43.6	48.0	55.1	61.4	69.8	74.0	76.4	74.6	71.9	64.0	52.5	47.7		
		MCDB	48.1	52.4	60.3	68.0	76.8	81.2	83.4	80.7	78.5	69.5	58.5	50.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	15.8	17.1	18.9	17.3	17.5	16.7	17.9	18.7	18.5	16.5	13.5		
		MCDBR	19.6	21.2	21.4	21.4	19.5	19.7	19.4	20.3	22.0	21.4	20.9	19.2		
		MCWBR	16.7	17.5	13.5	12.5	9.9	10.0	8.2	8.7	9.3	10.8	13.6	15.6		
	5% WB	MCDBR	18.9	20.6	19.3	18.6	17.2	17.3	16.8	16.4	18.4	17.5	18.1	16.8		
		MCWBR	18.0	18.5	13.3	12.5	9.2	9.5	8.6	8.4	8.9	10.2	14.4	15.4		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.322	0.351	0.400	0.432	0.444	0.461	0.460	0.418	0.355	0.323	0.301		
	taud		2.467	2.394	2.386	2.253	2.208	2.209	2.161	2.154	2.255	2.442	2.477	2.487		
	Ebn at Noon		276	286	287	278	270	266	260	257	261	270	267	270		
	Edh at Noon		25	31	35	42	45	45	47	46	39	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		588	827	1165	1529	1716	1976	1895	1755	1446	1042	703	495		
	RadStd		62	99	107	105	157	122	118	106	121	108	77	51		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	N/A

Nomenclature: See separate page